VIDYAPEETH

BATCH CODE: 12-AN-181-MA-2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:
UNITS & MEASUREMENT

Lecture No. 04

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Today's Goal

Subtopic .. Dimensional Analysis

Subtopic ..

Subtopic ..

Derived PQ :-

A PQ which can be explained using two or more fundamental PQs.

Eg :- $\text{Speed} = \dfrac{\text{length}}{\text{time}}$

length → Fund

time → F

ORDER OF MAGNITUDE :-

Order of Magnitude

Scientific Notation :-

$a \times 10^{b}$

$a < 10$

$a \leq 5$

$10 > a > 5$

Eg. $\underbrace{785.72}\times 10^{5}$

$= 7.8572 \times 10^{2} \times 10^{5}$

$$\boxed{x = 7.8572 \times 10^{7}}$$

52.2×10^{7}

$\downarrow$

$(5.22 \times 10) \times 10^{7}$

5.22×10^{8}

$a \times 10^b$

$a \leq 5$

$0.0.M = b$

3.14×10^7

$OOM = 7$

$10 > a > 5$

$0.0.M = b + 1$

9.14×10^7

$OOM = 7 + 1 = 8$

$F = 7.92 \times 10^{-34}$ N

SF & RO

a) 34

b) 35

c) 36

d) -33

DIMENSIONAL ANALYSIS

$x^1 + x^1 = 2x^1$

$F =$ mass $\rightarrow 1$
length $\rightarrow 1$
time $\rightarrow -2$

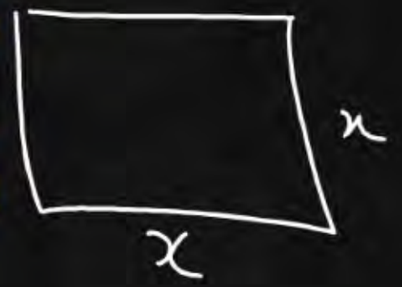
① $M^0 L^1 T^0$

$[L]$ length $= x$

$L \rightarrow$ dim $= 1$

② $M^0 L^2 T^0$

$[L^2]$



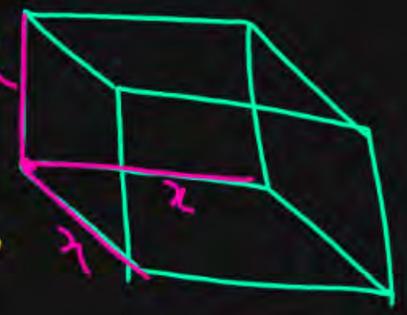
$A = x^2$
$= (\text{length})^2$

$L \rightarrow$ dim $= 2$

③

$[L^3]$

$M^0 L^3 T^0$



$\text{Vol} = x^3 = (\text{length})^3$

$L \rightarrow$ dim $= 3$

$F = \text{Kg} \cdot \text{m/s}^2 = M^1 L^1 T^{-2}$

$$\boxed{[F] = [MLT^{-2}]}$$

$$\text{Speed} = \frac{\text{Length}}{\text{time}} = \overset{\text{Unit}}{\frac{m}{s}} = [M^0 L^1 T^{-1} A^0 \theta^0 Cd^0 mol^0]$$

$$= [LT^{-1}]$$ Dimensional formula

$$= [M^0 L T^{-1}]$$

$$MLT^{-2}$$

PQ	Formula	SI unit	Dim. For
1. Vel	disp/time	m/s	$[M^0 L^1 T^{-1}]$
2. Accl	$\Delta v / t$	$\frac{m/s}{s} = m/s^2$	$[M^0 L^1 T^{-2}]$
3. "g" = Accl due to gravity		"	"
4. WORK	Force × displ $= (\text{Mass} \times \text{Accl}) \times \text{Disp}$ $= kg \cdot \frac{m}{s^2} \cdot m = kg\,m^2/s^2$		$[M^1 L^2 T^{-2}]$

5. Energy = Power × time

$= \frac{\text{Work}}{\cancel{\text{time}}} \times \cancel{\text{time}}$

$= \text{Force} \times \text{disp}$

$= \text{mass} \times \text{accl} \times \text{disp}$

$= kg \cdot \frac{m}{s^2} \cdot m = kg\, m^2/s^2 = [M^1 L^2 T^{-2}]$

$[\text{Energy}] = [M^1 L^2 T^{-2}]$

mass = 1

length = 2

time = −2

6. Current $[I] = [M L T A^1 \theta \, mol \, cd] = [A] = [M^0 L^0 T^0 A]$

7. Charge $\Rightarrow q = I \cdot t = [A \, T] = [M^0 L^0 T^1 A^1] = [q]$

8. Electric Potential $\Rightarrow V = \frac{W}{q} = [V] = \frac{M^1 L^2 T^{-2}}{T^1 A^1} = [M^1 L^2 T^{-3} A^{-1}]$

9. Electric Resistance $\Rightarrow R = \frac{V}{I} \Rightarrow [R] = \frac{M^1 L^2 T^{-3} A^{-1}}{A}$

$= [M^1 L^2 T^{-3} A^{-2}]$

10. Resistivity ; $\rho = \frac{RA}{L} = \frac{[\text{Resistance}][\text{Area}]}{[\text{length}]} = \frac{[M^1L^2T^{-3}A^{-2}][L^2]}{[L]}$

$[\rho] = [M^1L^3T^{-3}A^{-2}]$

11. [Universal grav. Const] = G $\quad 6.67\times10^{-11}\ \frac{Nm^2}{Kg^2}$

$F \propto m_1 m_2$

$F \propto \frac{1}{r^2}$

$$\boxed{F \propto \frac{m_1 m_2}{r^2}}$$

$F = G\frac{m_1 m_2}{r^2} \Rightarrow [G] = \left[\frac{Fr^2}{m_1 m_2}\right]$

$[G] = \frac{[MLT^{-2}][L^2]}{M^2}$

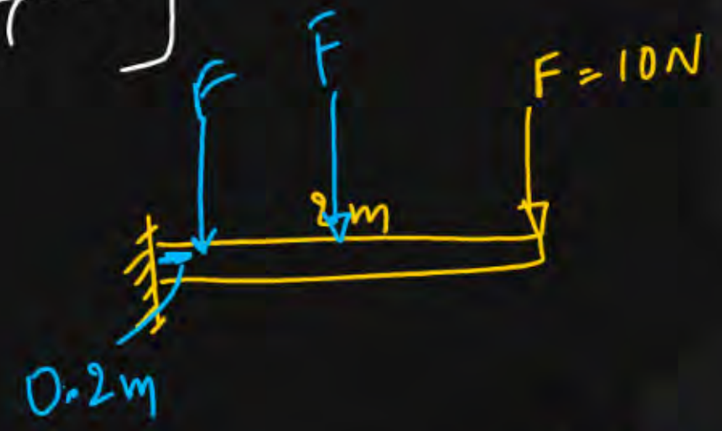
$[G] = [M^{-1}L^3T^{-2}]$

PQs having Same Dimensional formula :-

i) Work, Energy, Torque $= [M L^2 T^{-2}]$

Work (sc.)	Energy (sc.)	Torque (vector)
F.S	P.t	F × ⊥r dist
N m	N.m	$\vec{r} \times \vec{F}$
J	J	Nm

S.F $\xleftrightarrow{\text{Same}}$ F.S



Torque, $\tau = 20\,Nm$

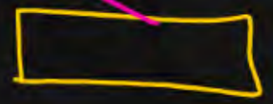
$\tau = 20\,Nm$

$\Rightarrow 20 = F \times r$

$\Rightarrow \frac{200}{0.2} = F = 100\,N$

2) Force, weight, Thrust $= [MLT^{-2}]$

Force



Gravitational force

$\downarrow$

$w = mg$

Perpendicular force

(3) Pressure, Stress $= [ML^{-1}T^{-2}]$

$\frac{\text{Ext. THRUST}}{\text{area}}$, $\frac{\text{Internal force}}{\text{area}}$

4. Momentum, Impulse $= [MLT^{-1}]$

$I = F \times dt$

$I = \frac{dP}{\cancel{dt}} \times \cancel{dt}$

$I = \frac{\Delta P}{\cancel{\Delta t}} \times \cancel{\Delta t}$

$\boxed{I = \Delta P}$

$r.i = \frac{\cancel{v_1}}{\cancel{v_2}}$

$= 1$

9.1×10^{-31}

1.67×10^{-27}

★★★★★

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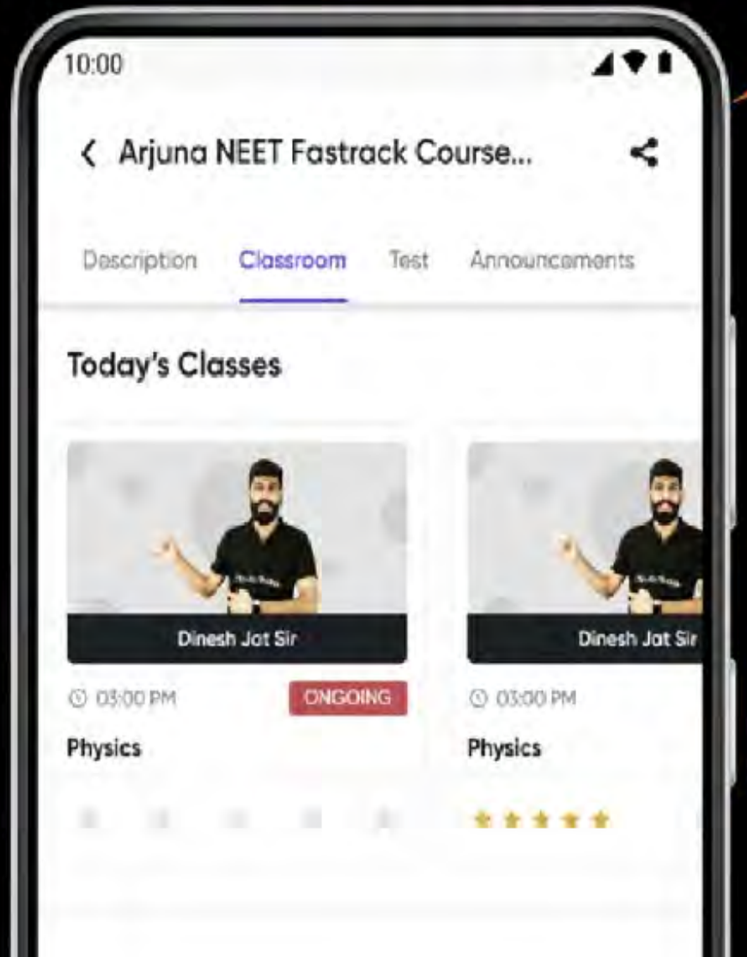
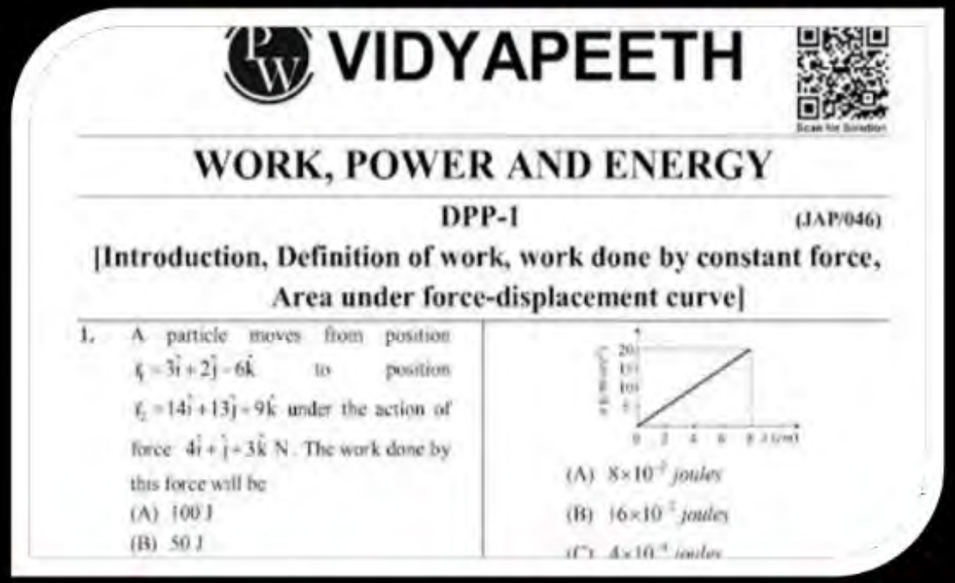


Solve the DPP and check Solution



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